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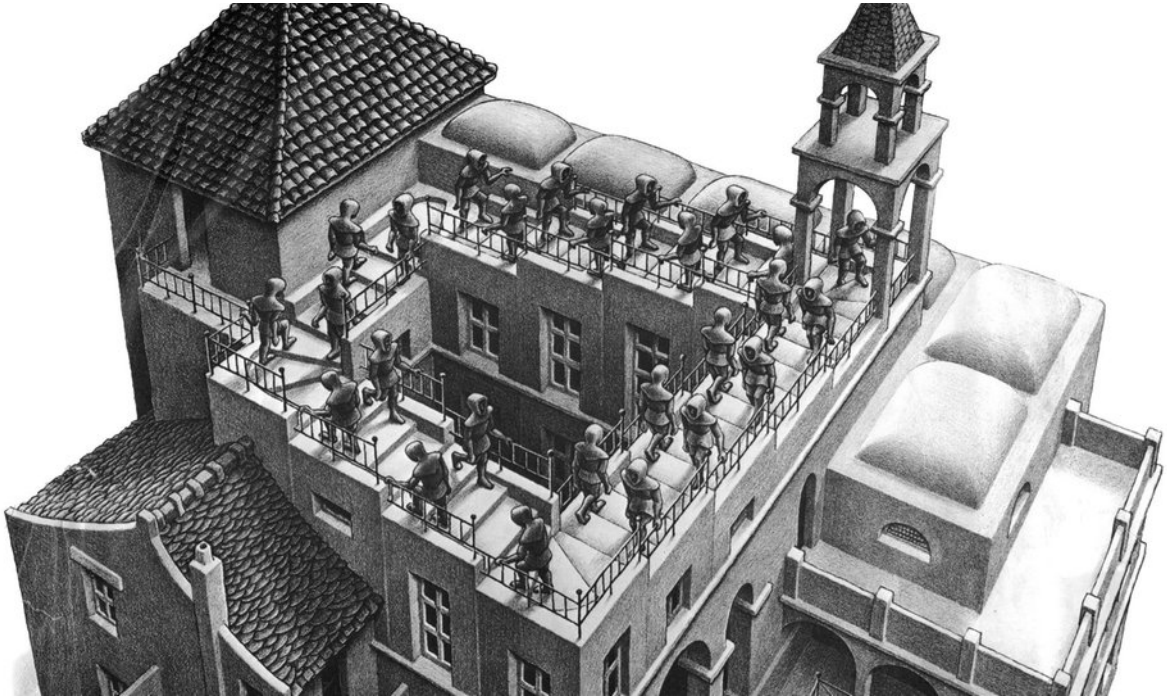
Software Engineering

Requirements Engineering

Leipzig, WS 26/27

Dr. Nico Willert Stark

LET`S TALK ABOUT REQUIREMENTS



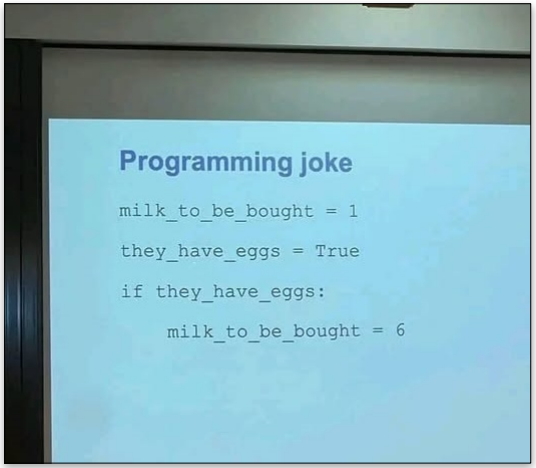
MC Escher, section of "Ascending and Descending, 1960

ARTIFICIAL INTELLIGENCE – THE ANSWER TO EVERYTHING!



Repost on X (Twitter) on 29.09.2024 by Allen Holub (@allenholub / @allenholub.bsky.social)

ANALYSIS ERROR – WHAT COULD GO WRONG?



My wife said: „Please go to the store and buy a carton of milk and if they have eggs, get six.“

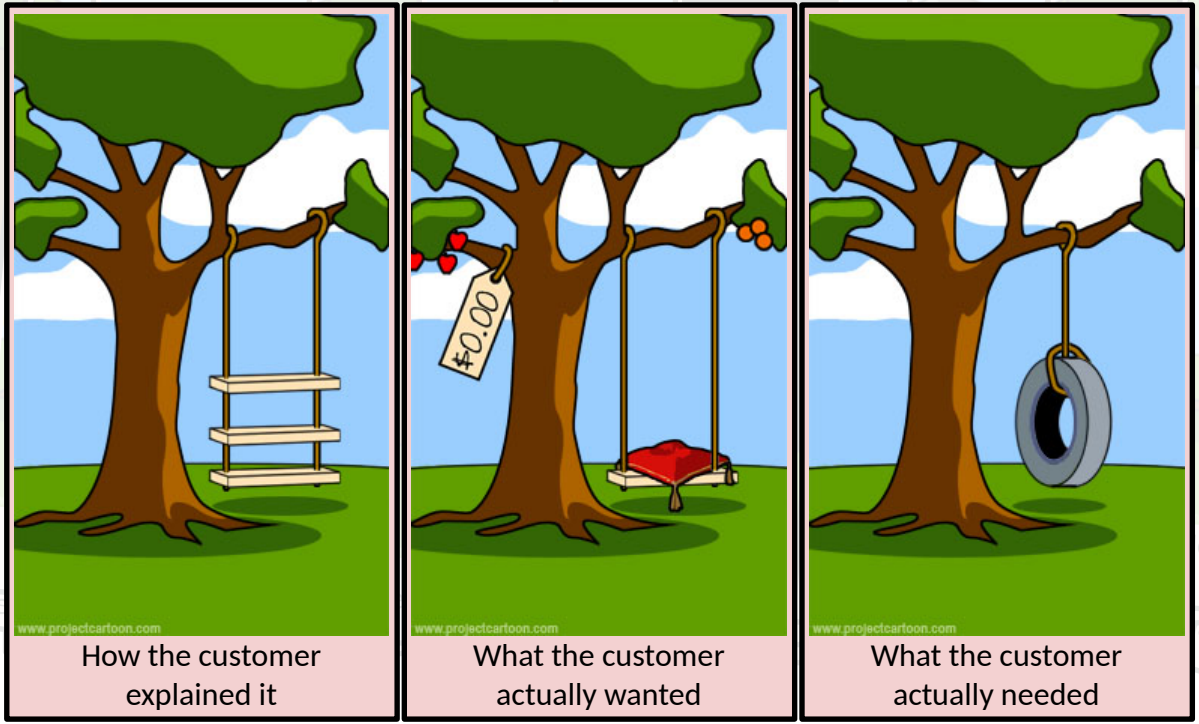
I came back with 6 cartons of milk.

She said: „Why in the hell did you buy six cartons of milk?“

I said: „They had eggs“

Quelle: Facebook, Programmer Humor site, retrieved on 06.12.2024

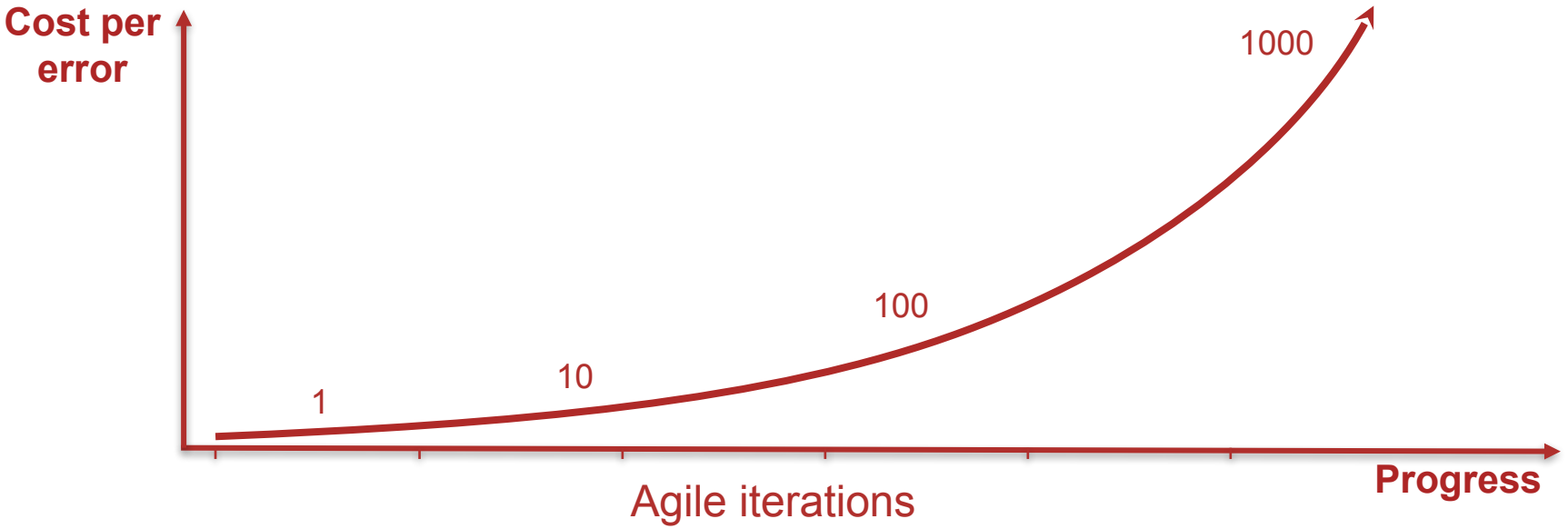
Discrepancies can be minimized by targeted and correct Requirements Management



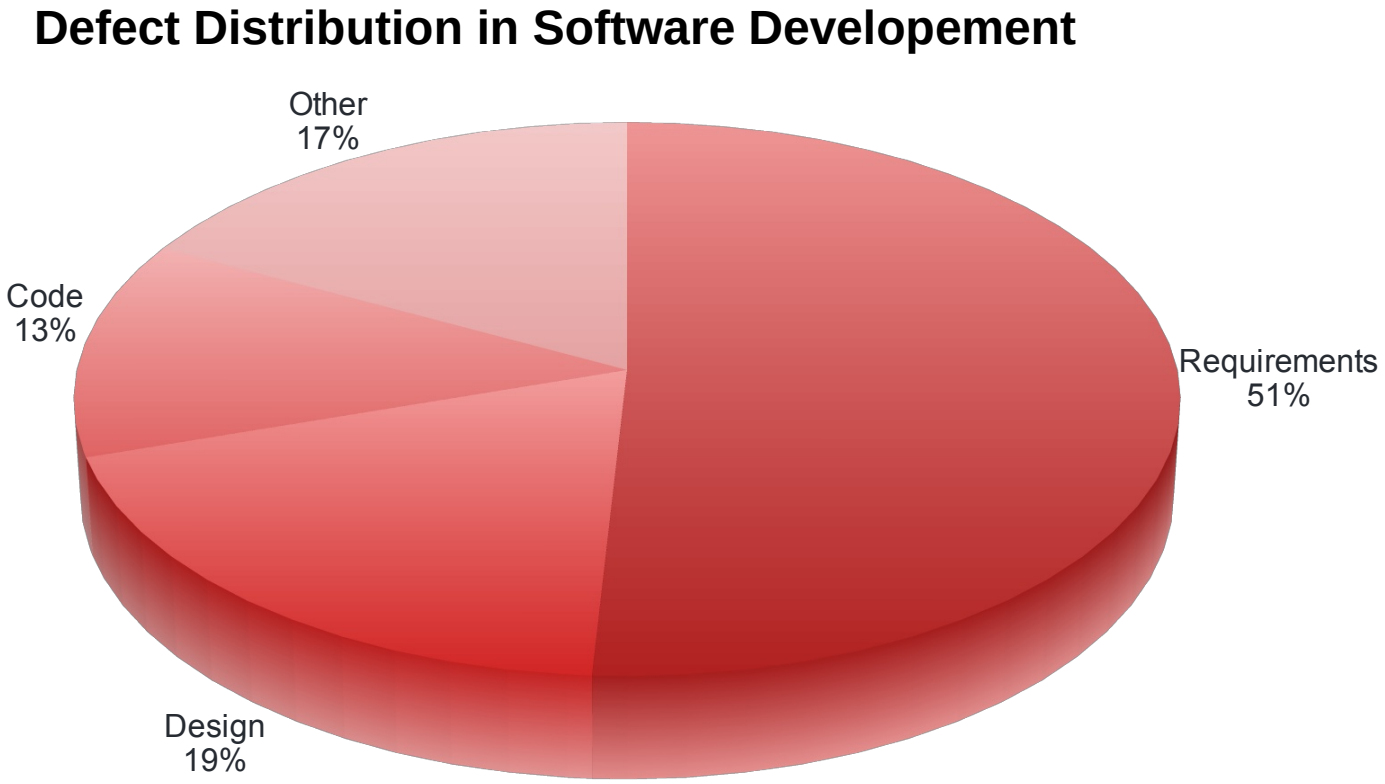
How the customer explained it	What the customer actually wanted	What the customer actually needed

http://projectcartoon.com

THE COST OF ERRORS



Own presentation according to T. Pfeifer, Qualitätsmanagement, 2. Aufl., 1996, p. 11. The corresponding presentation can be found in Pfeifer/Schmitt 2010 on p. 154.



Software Quality Through Requirement and Design, Massood Towhidnejad, Embry-Riddle Aeronautical University, 2006

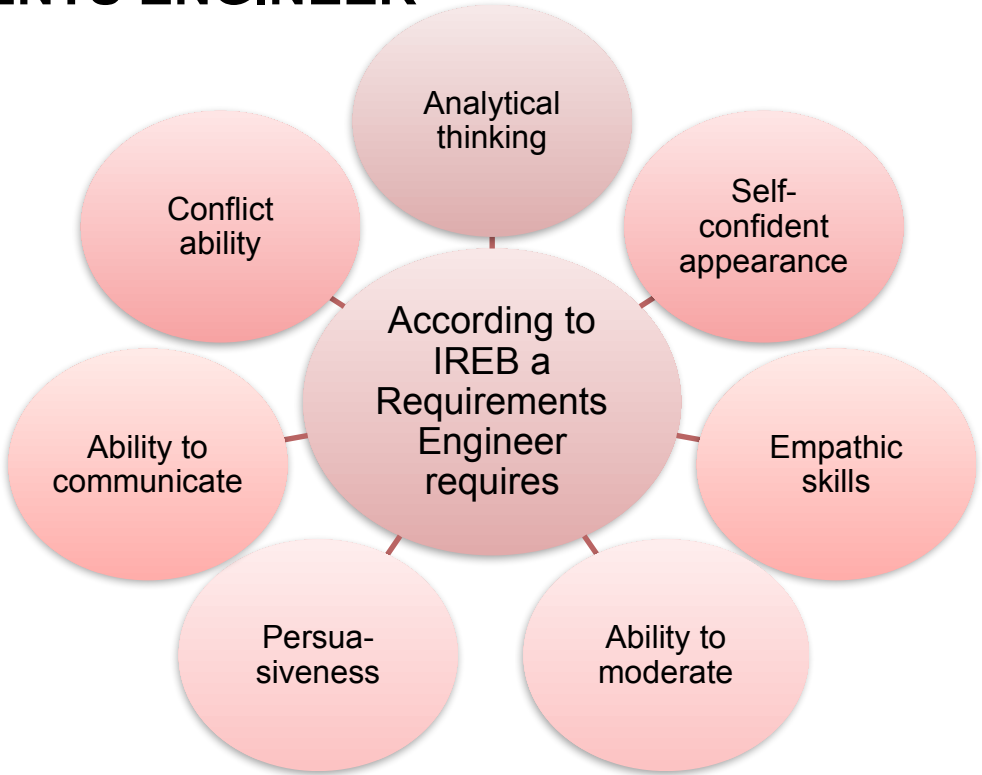
DEFINITION REQUIREMENTS ENGINEERING (IREB*)

- Requirements engineering is a **cooperative**, **iterative**, **incremental process** that aims to ensure that:
  - **All relevant requirements are known** and software engineers should understand the **level of detail required**
  - The **stakeholders** involved **achieve sufficient agreement** on the known requirements
  - All **requirements are documented** in accordance with the **documentation regulations** or specified in accordance with the specification regulations

*IREB: International Requirements Engineering Board

(Klaus Pohl, Requirements Engineering, 2010)

REQUIREMENTS ENGINEER



- Ichbewusstsein
- Kontextbewusstsein
- Motivierendes Wesen
- Führungspersönlichkeit
- Flexibilität
- Reflexion
- Neutralität
- Interkulturelle Kompetenzen
- Ethisches Gewissen

(Rupp & Die SOPHISTen, 2021, p. 13)

DEFINITION OF REQUIREMENTS ACCORDING TO IEEE AND IREB

- A requirement is
  1. A **condition or capability** required by a user (person or system) to **solve a problem or achieve a goal**
  2. A condition or capability that a **system or subsystem must meet or possess** in order **to meet a contract, standard, specification or other formally specified document**
  3. A **documented representation** of a condition or property according to (1) or (2)
- Or simplified
  - A requirement is a **statement about a characteristic or performance** of a **product**, a **process** or the **persons involved** in the process

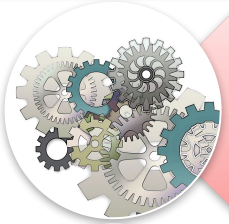
*IEEE: Institute of Electrical and Electronics Engineers

(Rupp & Die SOPHISTen, 2009, pp. 12-13)

REQUIREMENTS (1/3)



However, a requirement is **not only a request for a service** that was provided at the end of software development



There are requirements in different **degrees of detail and** in different **classifications**



There are requirements for **software**, **hardware**, **test cases**, **manuals**, **protocols**, **safety clearance** of employees, certain **qualifications** of developers

(Rupp & Die SOPHISTen, 2009, p. 14)

REQUIREMENTS (2/3)



Requirements must reflect the **common understanding** and knowledge of those involved



Requirements are the basis for **invitations to tender**, **contract designs** and system **approvals**

Comparison ACTUAL (how does the system actually react) with TARGET (what define the requirements)

A precise definition of the requirements is essential*.

*generally speaking…

(Rupp & Die SOPHISTen, 2009, p. 16)

REQUIREMENTS (3/3)

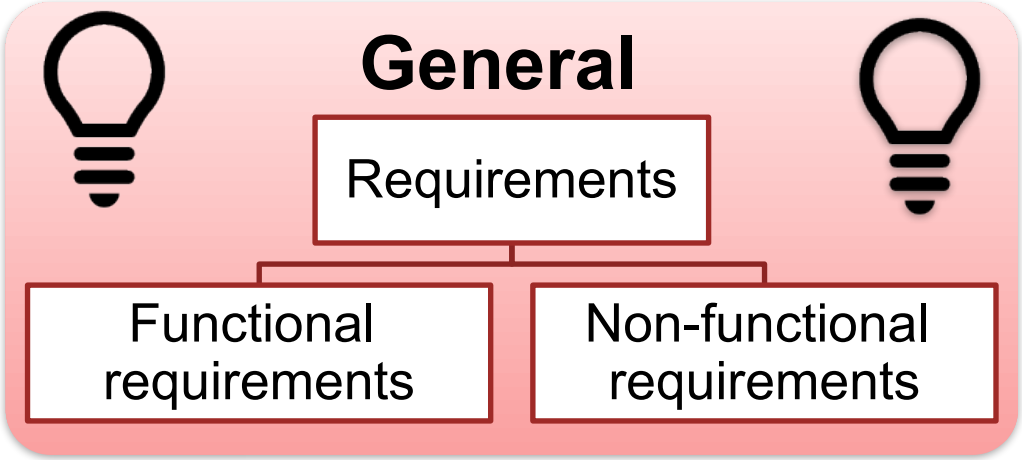
Activities and their methods in requirements engineering

Collection	Interview
	Questionnaire
Documentation	Use cases
	Style guides
Validation	Review
	Test case creation
Managing	Releases
	Branches

(Rupp & Die SOPHISTen, 2009, p. 14)

CLASSIFICATION OF REQUIREMENTS (1/2)

- IREB
  - Functional requirements
  - Quality requirements
  - **Boundary conditions**
- Or others say
  - Functional requirements
  - Technological requirements
  - Quality requirements
  - Requirements for the user interface
  - Requirements for other delivery components
  - Requirements for activities to be performed and legal-contractual requirements



(Rupp & Die SOPHISTen, 2009, p. 15)

CLASSIFICATION OF REQUIREMENTS (2/2)

Functional requirements	Non-functional requirements
Describe desired functionalities • "The product **shall** be able to make coffee." • "The product **should** have an on/off switch."	**Describe requirements for "quality"** • "The coffee **shall** be ready within 30 seconds." • "The product **should** run 24x7."
Client will mainly formulate functional requirements	The client will probably only think of non-functional requirements if he has had bad experiences in the past (e.g., poor quality, low availability)
Differ from product to product	Can be similar or even standardized for many of the company's products, e.g., a uniform appearance
Usually have only "local" effects on the product	Usually have "global"/far-reaching impact on product

CLASSIFICATION BY LIABILITY

Liability	German keyword	English keyword	Legal validity
Duty	**muss**	**shall**	Legally binding - mandatory to fulfil
Wish	**sollte**	**should**	Not legally binding - but indicates the intention
Purpose	**wird**	**will**	Indicates future developments that must be taken into account – can be legally binding



Or shall, should, may, can, will



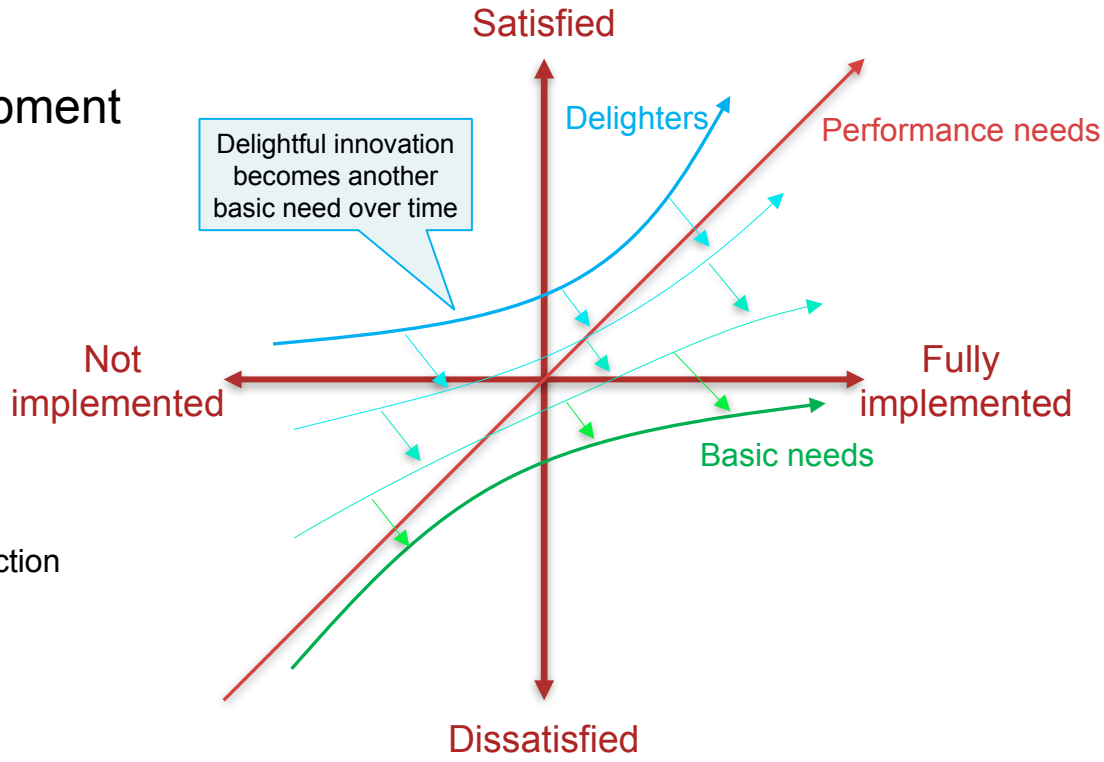
If necessary, also define non-requirements.

a 100% right or wrong does not exist, but should be considered in the contract

(Rupp & Die SOPHISTen, 2014, pp. 18-19)

KANO MODEL (1/2)

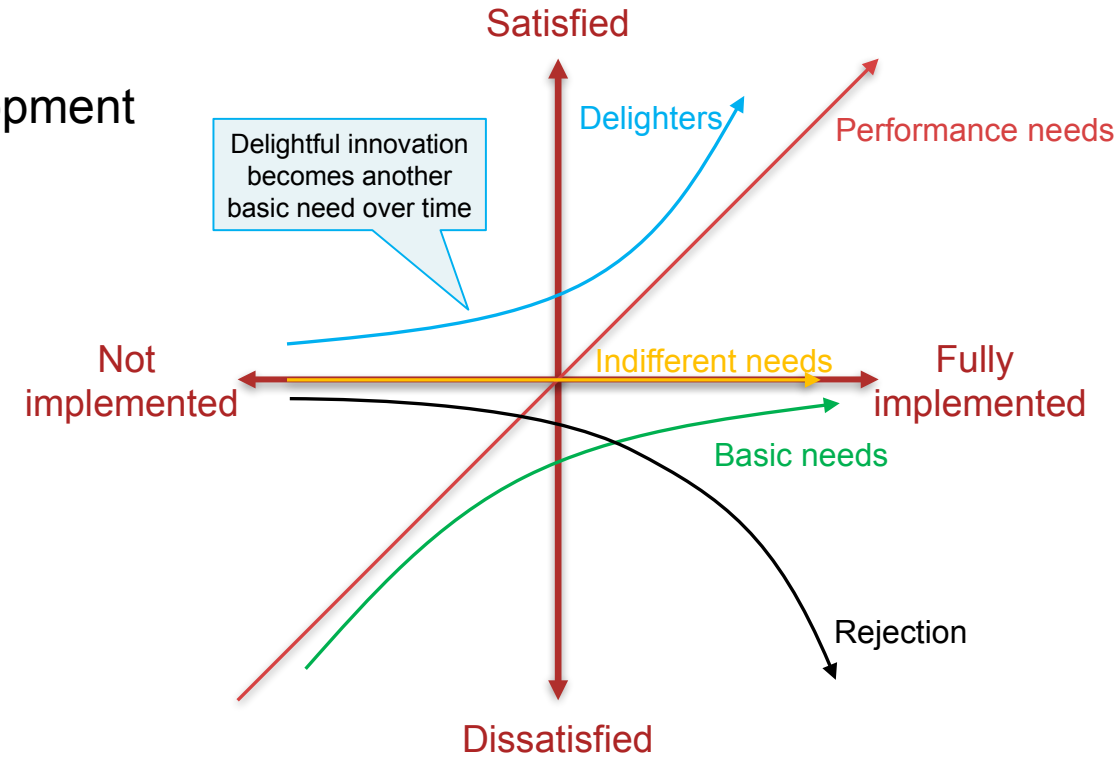
- Is a theory for product development and **customer satisfaction**
  - Basic needs / Must-be
    - Customer expects
    - Well done → neutral
    - Poorly done → dissatisfaction
  - Performance needs
    - Customer is aware of
    - Well done → satisfaction
    - Not/poorly fulfilled → dissatisfaction
  - Delighters / Attractive
    - Customer doesn't expect
    - Well done → satisfaction
    - Not/poorly fulfilled → neutral



(Rupp & Die SOPHISTen, 2014, pp. 95-97; Page "Kano-Modell". In: Wikipedia, The Free Encyclopedia https://de.wikipedia.org/w/index.php?title=Kano-Modell&oldid=180219516 (Retrieved: October 17, 2018))

KANO MODEL (2/2)

- Is a theory for product development and customer satisfaction
  - Indifferent needs
    - Customer doesn't care
    - Well done → neutral
    - Poorly done → neutral
  - Rejection / reverse needs
    - Customer doesn't want this
    - Implemented→ dissatisfaction
    - Not implemented→ neutral



(Rupp & Die SOPHISTen, 2014, pp. 95-97; Page "Kano-Modell". In: Wikipedia, The Free Encyclopedia https://de.wikipedia.org/w/index.php?title=Kano-Modell&oldid=180219516 (Retrieved: October 17, 2018))

REQUIREMENTS

Main problems of requirements analysis

- Unclear objectives
- High complexity
- Language barriers
- **Poor quality**
- Volatile requirements
- Loss of information

Requirements for requirements

or: How can I tell whether a requirement is well formulated?

- Complete
- Necessary
- Atomic
- Traceable
- Technically solution-neutral
- Realizable
- Consistent
- Unambiguous
- Testable

(Rupp & Die SOPHISTen, 2014, pp. 24-28)

POORLY FORMULATED REQUIREMENT

Formulate liability
shall, should, may, can, will

Avoidance of indefinite quantifiers
Bad: any, most of the time, usual, somehow
Better: all, everyone, and, always, or, none

It would be nice if the data could somehow simply be exported to Excel using json2xls.

Completely specify nouns
Which data?

Complete comparisons and increases
Simply to perform in comparison to what for?

Do not specify any solutions

(Belikan, 2007)

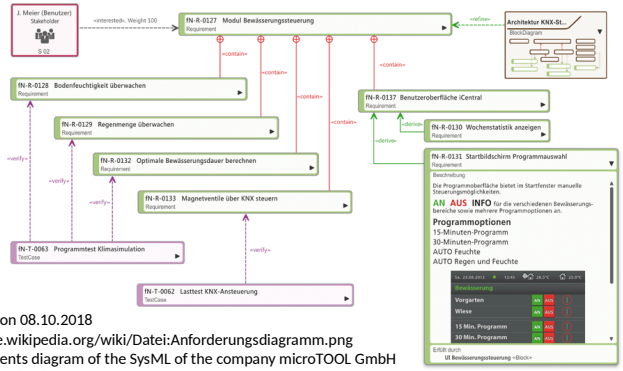
REQUIREMENTS

Main problems of requirements analysis

- Unclear objectives
- **High complexity**
- Language barriers
- Poor quality
- Variable requirements
- Loss of information → documentation!

There are connections between different requirements

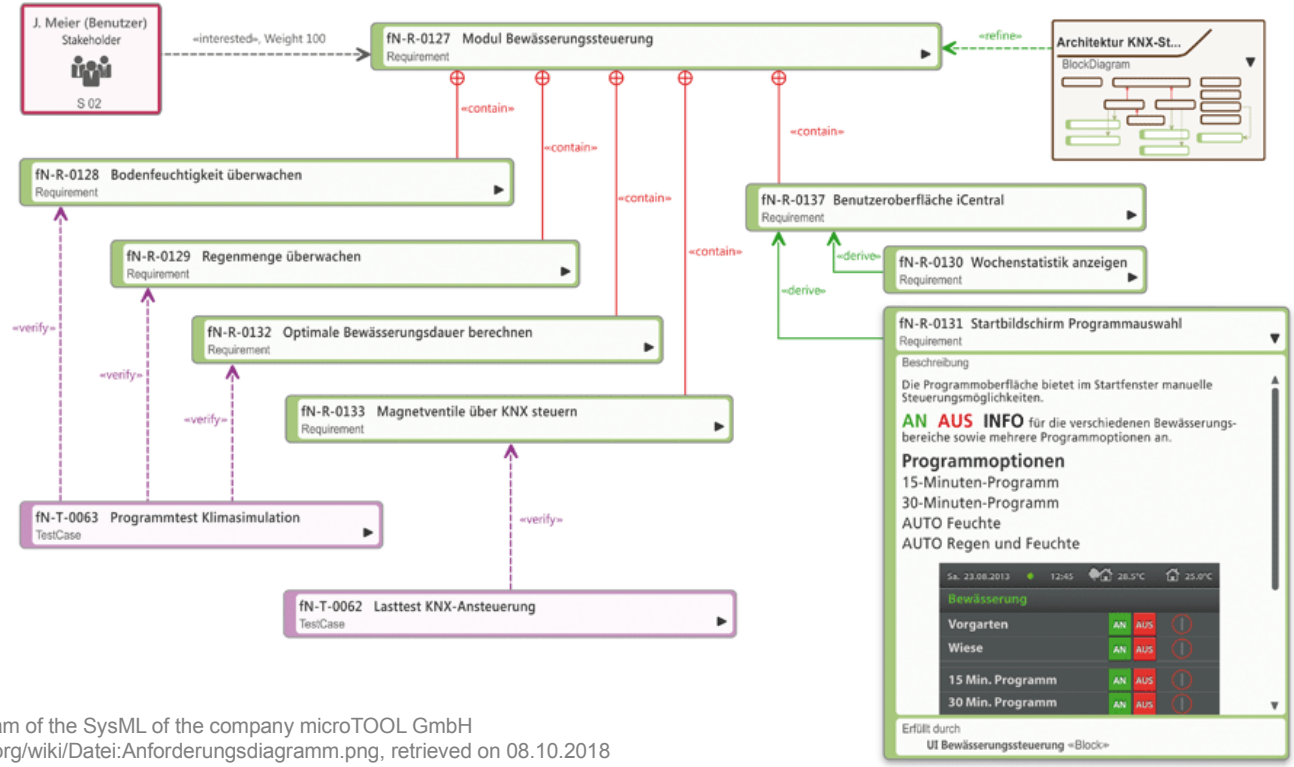
- Especially when requirements are specifications of other requirements



retrieved on 08.10.2018
https://de.wikipedia.org/wiki/Datei:Anforderungsdiagramm.png
Requirements diagram of the SysML of the company microTOOL GmbH

(Rupp & Die SOPHISTen, 2014, pp. 36-39)

REQUIREMENTS DIAGRAM



Requirements diagram of the SysML of the company microTOOL GmbH
https://de.wikipedia.org/wiki/Datei:Anforderungsdiagramm.png, retrieved on 08.10.2018

5 SPECIFICATION LEVELS

0

- Very abstract requirement
- Describes the total scope in a few sentences
- To give an overview, Introduction
- Contains declarations of intent, objectives, main features or the “mission statement”

Management “Big Picture”

1

- Identifying requirements sub-steps of the business process
- Description with use cases, epics

Department employees must understand these requirements

2

- Core of the system analysis
- Technical processes are defined for processes from specification level 1.
- Tasks arise for each of the parties involved.
- User stories

Department employees must understand the requirements of the processes that affect them

3

- Forming the conclusion of the technical specifications
- Consider the (sub)processes at level 2 in more detail and refine them.

Department employees must understand the requirements of the processes that affect them

4

- Considers the separation of the system into software and hardware or other components
- Requirements from these system components
- Offers technological solutions for requirements of other specification levels

They are no longer relevant for department employees, the architect must understand them

(Rupp & Die SOPHISTen, 2014, pp. 40-41)

PREPARATION

- Basis must be created before requirements are established
  - Develop knowledge about context
  - Define limits of the system to be developed
  - Define the objectives of the project
  - **Determine sources of requirements**
    - **Stakeholders**
    - **Documents (e.g., legislative texts)**
    - **Risk analysis**
    - **Product ideas**
    - **Best practice**
    - **Other systems (,e.g., predecessor or competitor systems)**



(Rupp & Die SOPHISTen, 2014, p. 74)

EXAMPLE LEGISLATIVE TEXTS

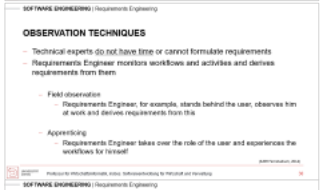
STRAßENVERKEHRS-ZULASSUNGS-ORDNUNG (STVZO)
§ 53 SCHLUSSLEUCHTEN, BREMSLEUCHTEN, RÜCKSTRAHLER

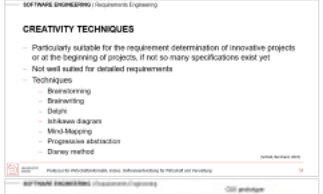
- (1) Kraftfahrzeuge und ihre Anhänger müssen hinten mit **zwei** ausreichend wirkenden Schlussleuchten für **rotes Licht** ausgerüstet sein. [..] Der **niedrigste Punkt der leuchtenden Fläche der Schlussleuchten darf nicht tiefer als 350 mm**, [..] und der **höchste Punkt der leuchtenden Fläche nicht höher als 1 500 mm**, [..]

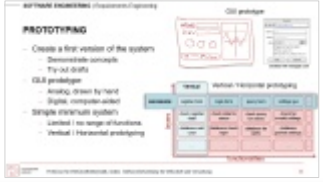
INVESTIGATIVE TECHNIQUES FOR REQUIREMENTS

- Interview techniques
  - Interview
  - Questionnaire
- Observation techniques
  - Field observation
  - Apprenticing
- Creativity techniques
  - Brainstorming (written/verbal)
  - Workshop
- Prototyping
  - GUI-prototype











INTERVIEW TECHNIQUES

- Stakeholders must have time and be able to express requirements
- Suitable for all types of requirements
- Interview
  - Prepare, ask and directly log questions, if necessary extra logger
- Questionnaire

OBSERVATION TECHNIQUES

- Technical experts do not have time or cannot formulate requirements
- Requirements Engineer monitors workflows and activities and derives requirements from them

  - Field observation
    - Requirements Engineer, for example, stands behind the user, observes him at work and derives requirements from this

  - Apprenticing
    - Requirements Engineer takes over the role of the user and experiences the workflows for himself

(IUBH Fernstudium, 2014)

CREATIVITY TECHNIQUES

- Particularly suitable for the requirement determination of innovative projects or at the beginning of projects, if not so many specifications exist yet
- Not well suited for detailed requirements
- Techniques
  - Brainstorming
  - Brainwriting
  - Delphi
  - Ishikawa diagram
  - Mind-Mapping
  - Progressive abstraction
  - Disney method

(Schloß, Bernhard, 2015)

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- Group of any number of persons
  - Experts, employees, laymen ...
- Collecting ideas
  - No criticism, evaluation or judgement of the ideas
  - No killer phrases
  - Be imaginative
- Sorting and evaluating ideas

Page “Brainstorming". In: Wikipedia, The Free Encyclopedia
https://de.wikipedia.org/w/index.php?title=Brainstorming&oldid=176567551 (Retrieved: October 10, 2018)

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- Similar to brainstorming
- Ideas will be written by each participant, anonymously if necessary
- Several rounds: Write down ideas, collect them, present them free of criticism
- Subsequently evaluate results in detail

Page "Brainwriting". In: Wikipedia, The Free Encyclopedia
https://de.wikipedia.org/w/index.php?title=Brainwriting&oldid=180833910 (Retrieved: October 10, 2018)

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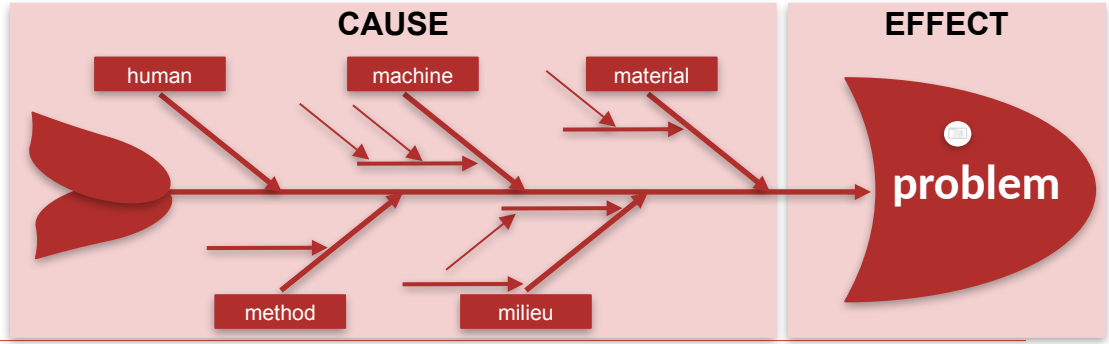
- Multi-stage expert survey
- Experts evaluate given theses
  - Estimates are presented anonymously with justification
  - Several passes, opinions are getting more and more approximated

Page "Delphi-Methode". In: Wikipedia, The Free Encyclopedia
https://de.wikipedia.org/w/index.php?title=Delphi-Methode&oldid=179029940 (Retrieved: October 10, 2018)

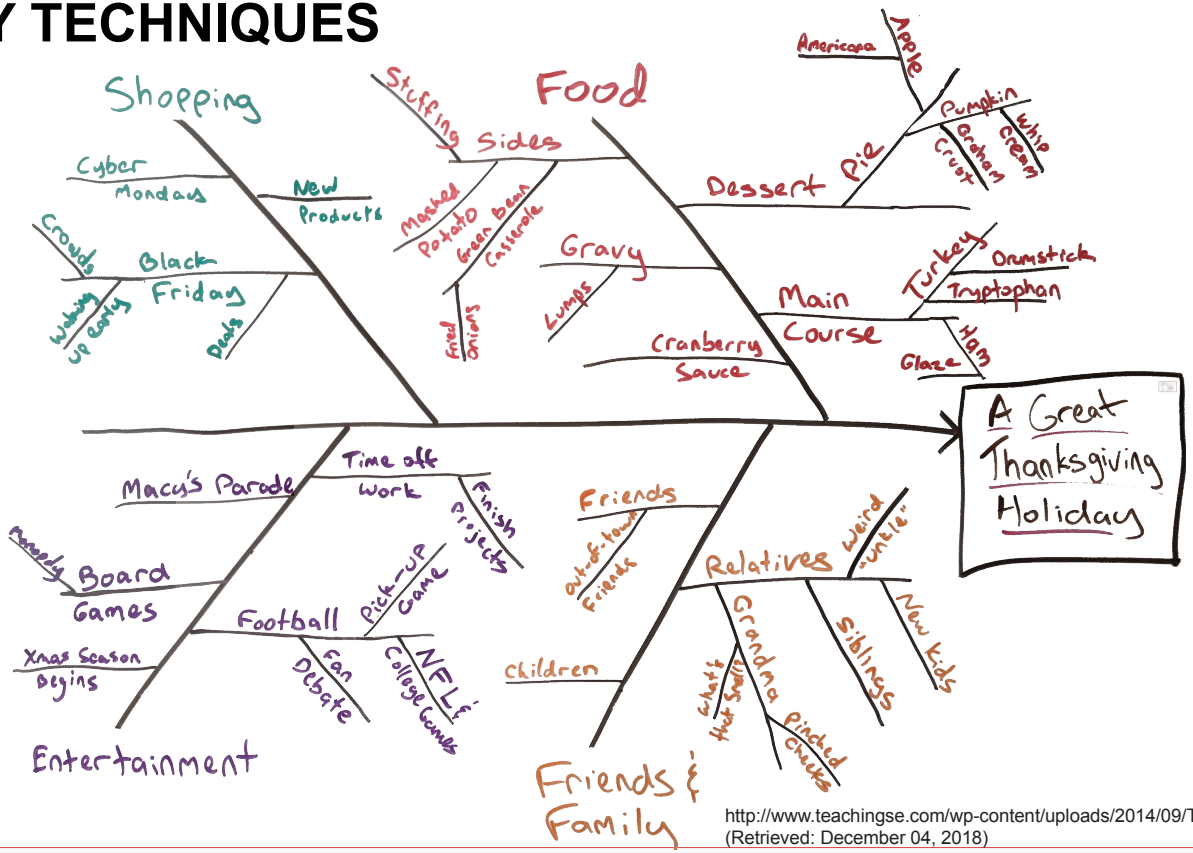
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  - **Ishikawa diagram**
  - Mind-Mapping
  - Progressive abstraction
  - Disney method

- Investigates causes in different dimensions
  - human, machine, material, milieu, management, method, measurement, money…



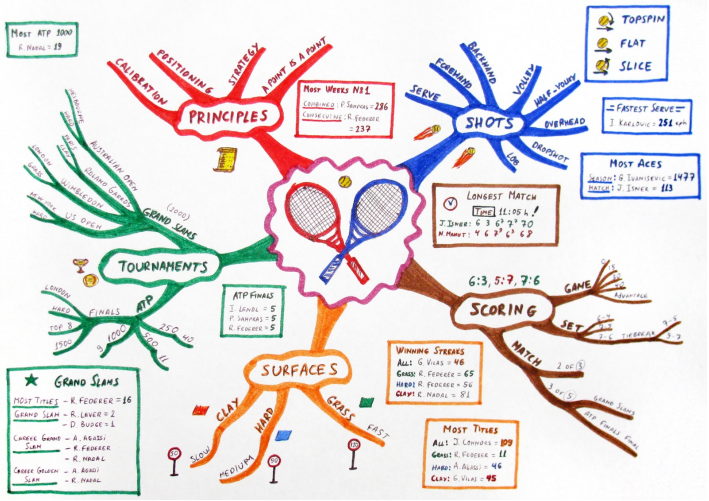
CREATIVITY TECHNIQUES



http://www.teachingse.com/wp-content/uploads/2014/09/ThanksgivingFishbone1.gif
(Retrieved: December 04, 2018)

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https://upload.wikimedia.org/wikipedia/commons/1/1a/Tennis-mindmap.png (Retrieved: October 10, 2018)

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  - Mind-Mapping
  - **Progressive abstraction**
  - Disney method

- New solutions can be found by changing the perspective of the project
  - Find problem at next level up
    - "What's the big deal?"
    - "How else could this work?"
  - If no solution is found, abstract further
- Reverse psychology (German: Paradoxe Intervention)
  - "What can we do to make the problem worse?"

Page "Progressive Abstraktion". In: Wikipedia, The Free Encyclopedia
https://de.wikipedia.org/w/index.php?title=Progressive_Abstraktion&oldid=169414231 (Retrieved: October 10, 2018)

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  - Progressive abstraction
  - **Disney method**

- "…there were actually three different Walts: the dreamer, the realist, and the spoiler"

  Robert B. Dilts
- Problem is viewed from 3 angles
  - The dreamer: subjective and enthusiastic
  - The realist: pragmatic-practical
  - The critic: searching and examining
  - Can be used by individuals or groups

Page "Walt-Disney-Methode". In: Wikipedia, The Free Encyclopedia
https://de.wikipedia.org/w/index.php?title=Walt-Disney-Methode&oldid=181458213 (Retrieved: October 10, 2018)

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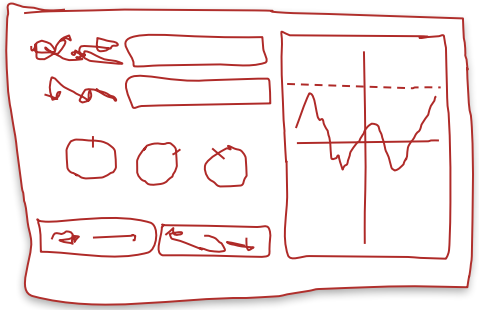


(Schloß, Bernhard, 2015)

PROTOTYPING

- Create a first version of the system
  - Demonstrate concepts
  - Try out drafts
- GUI prototype
  - Analog, drawn by hand
  - Digital, computer-aided
- Simple minimum system (MVP)
  - Limited / no range of functions
  - Vertical / Horizontal prototyping

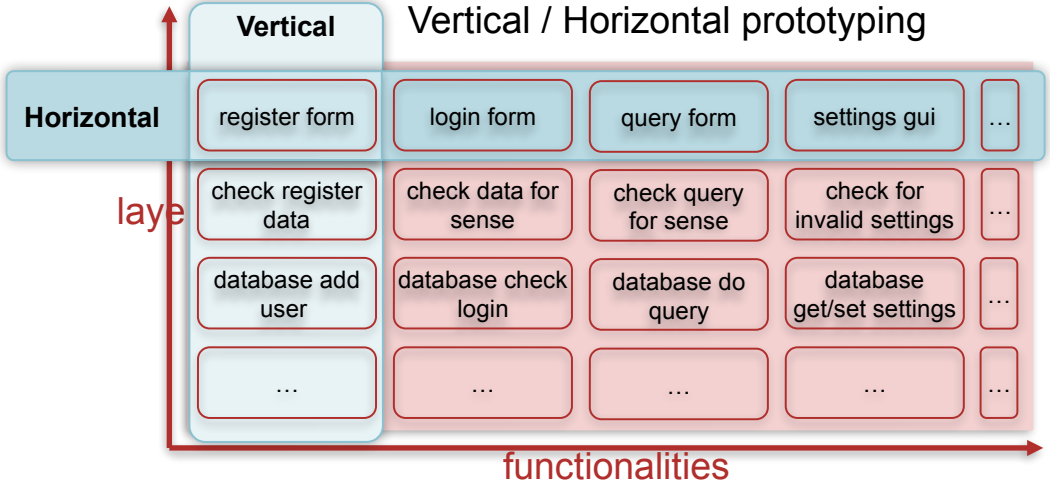
GUI prototype





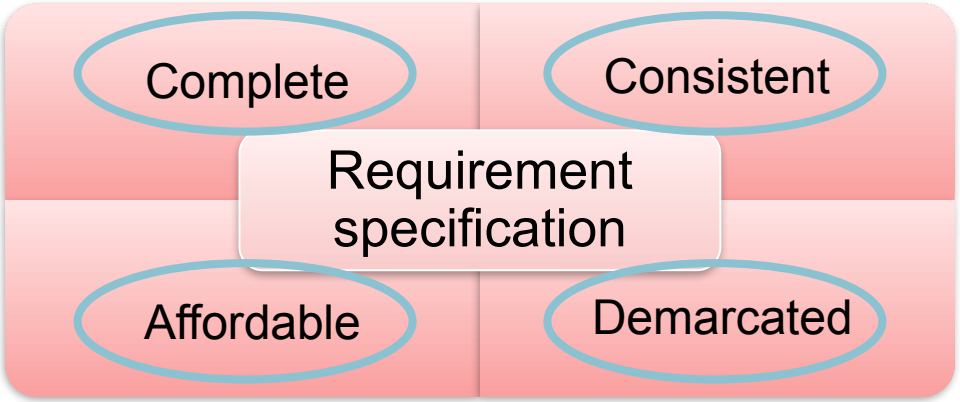
created with moqups.com

Vertical / Horizontal prototyping



REQUIREMENT SPECIFICATION

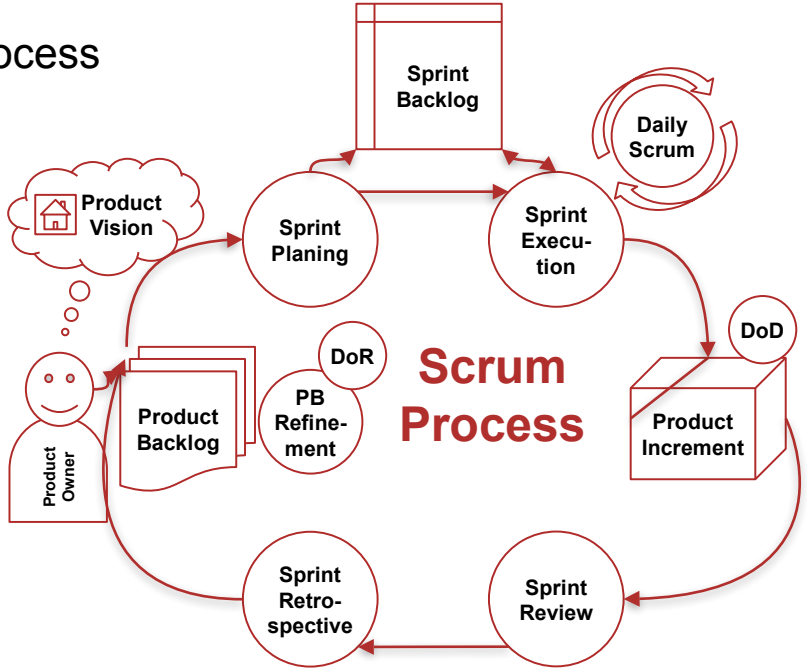
- The totality of all requirements is called requirement specification. This must also meet certain quality criteria.



(Rupp & Die SOPHISTen, 2014, pp. 28)

IMPLEMENTATION IN SCRUM

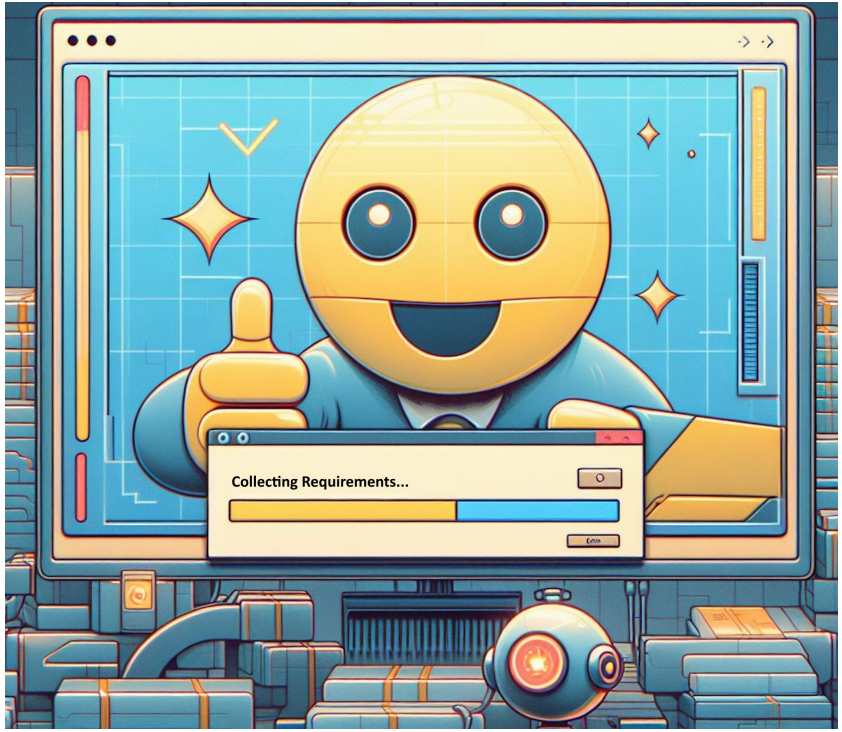
- RE implementation not immediately apparent, Process does not provide role of Requirements Engineer
  - Product Owner and Development Team
    - Product Backlog Sprint Backlog
- Reminder: Product Backlog
  - Product vision - Specification level 0
  - Themes / Epics - Specification level 1
  - User Stories - Specification level 2
    - Acceptance Criterias
- Definition of Ready + Definition of Done
  - Can be used for quality criteria of user stories and product increments



(Rupp & Die SOPHISTen, 2014, pp. 59-63)

AI SUPPORT OF REQUIREMENTS ENGINEERING

- **Analyze** existing source code, user interface screenshots or documentation (NLP).
- **Gather** requirements from unstructured data, e.g. user feedback, customer reviews, speech or language translation.
- **Generate** visualizations, source code, user interfaces or documentation.
- **Clarify and validate** requirements.
- **Assess the quality** of requirements, e.g. identify inconsistencies and conflicts.
- **Simulate** system behavior.
- **Track changes** to requirements.
- **Manage dependencies** between requirements.



REFERENCES



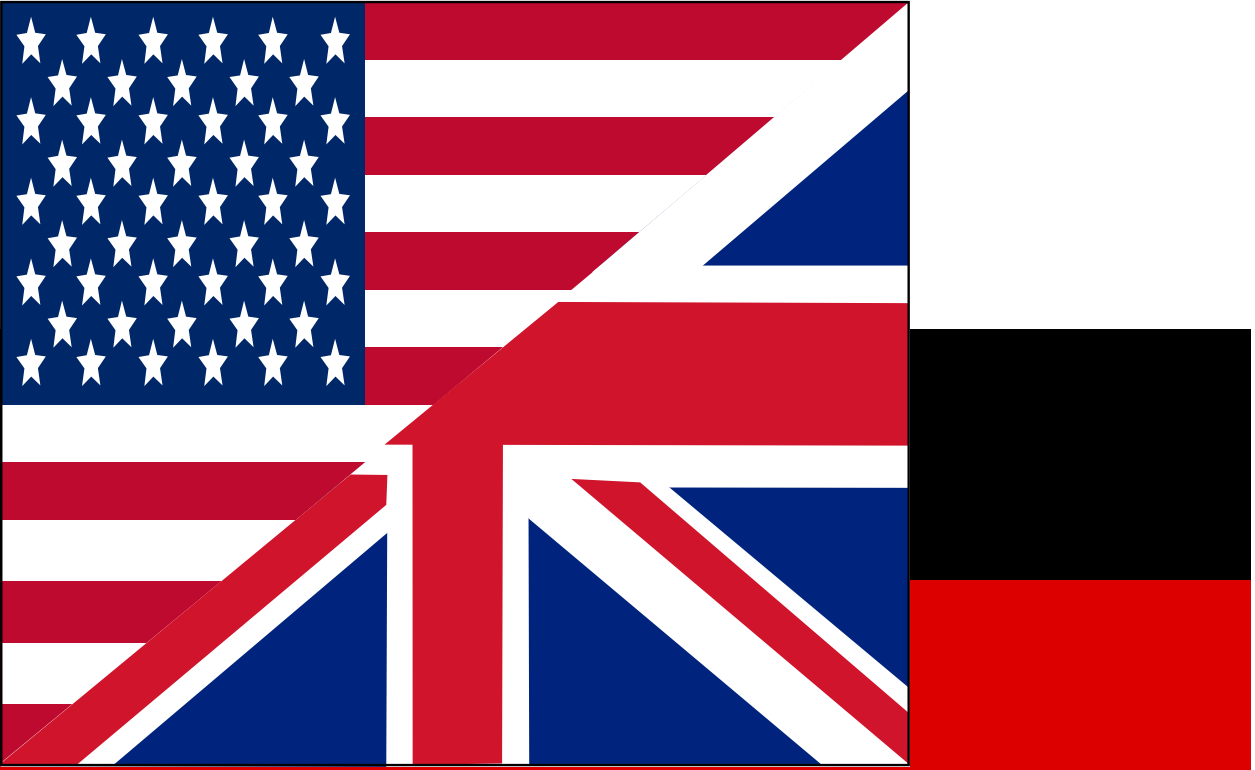
By Stewart Butterfield - flickr, CC BY 2.0, https://commons.wikimedia.org/w/index.php?curid=1093147

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https://openclipart.org/detail/168121/usuk-flag
https://openclipart.org/detail/296539/german-flag

GLOSSARY (1)

- affordable: erschwinglich
- apprenticing: Lehrlingsausbildung
- approximate : annähernd
- boundary condition: Rahmenbedingung
- competitor: Konkurrent
- contract design: Vertragsgestaltung
- demarcated: abgegrenzt
- duty: Pflicht
- field observation: Feldbeobachtung
- invitation to tender: Ausschreibung
- justification: Begründung
- liability: Verbindlichkeit
- measurement: Messung
- persuasiveness: Überzeugungskraft
- predecessor: Vorgänger
- questionnaire: Fragebogen
- requirement: Anforderung

GLOSSARY (2)

- requirements engineering: Anforderungsanalyse
- requirements management: Anforderungsmanagement
- scope: Umfang
- spoiler: Spielverderber
- stakeholder: Interessenvertreter
- system approval: Systemabnahme
- unambiguous: unmissverständlich / eindeutig
- workflow: Arbeitsablauf
- acceptance criterias: Akzeptanzkriterien

YOUR QUESTIONS, PLEASE?



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THANK YOU VERY MUCH!

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